

STATE-SELECTED FINITE DIFFERENCES: COMPLETE FIBRES AND JORDAN BLOCKS

ANONYMOUS

ABSTRACT. For a prime p , consider all functions $f : \mathbb{F}_p \rightarrow \mathbb{F}_p$ and update

$$f(x) \mapsto f(x + f(0)) - f(x).$$

The current value at zero selects a new translation direction at every epoch, so the map is nonlinear despite being assembled from linear differences. We prove a complete finite atlas. The t th image is the t th cyclic-difference layer and has size p^{p-t} for $0 \leq t \leq p$; each nonzero target in that layer has exactly $(p-1)^t$ time- t preimages, while the zero fibre is obtained explicitly by mass balance. The functional graph is one rooted component of sharp height p , with closed depth shells. Its complex transition operator has one $J_1(1)$ block, $(p-1)^2 p^{p-s-1}$ blocks $J_s(0)$ for $s < p$, and $p-1$ blocks $J_p(0)$. Fixed finite differences, augmentation-ideal nilpotence, and generic rank-to-Jordan conversion are assigned no contribution credit. The literal feedback conjunction remains owner-thin and externally held.

1. THE FEEDBACK MAP AND THEOREM

Fix a prime p and put

$$\mathcal{V} = \mathbb{F}_p^{\mathbb{F}_p} = \{f : \mathbb{F}_p \rightarrow \mathbb{F}_p\}.$$

For $a \in \mathbb{F}_p$, let $(\tau_a f)(x) = f(x + a)$ and $D_a = \tau_a - I$. Our literal self-map is

$$(1) \quad T(f) = D_{f(0)} f.$$

Thus $T(f) = 0$ immediately when $f(0) = 0$; otherwise the direction used at the next epoch depends on the new state. Write $\tau = \tau_1$, $N = \tau - I$, and

$$J^t = N^t \mathcal{V} \quad (0 \leq t \leq p).$$

Difference operators and their augmentation-ideal formulation are standard [1]. The nilpotent/bijective decomposition and functional graphs of fixed linear maps over finite fields are likewise classical [2]. We assign the filtration $J^\bullet$, fixed-difference nilpotence, affine kernel counts, and rank-to-Jordan conversion no contribution credit. The object isolated here is the state-selected direction in (1), especially its anchored inverse system.

Theorem 1 (complete temporal, inverse, and algebraic atlas). *For every prime p , the map (1) has the following properties.*

(i) *For $0 \leq t \leq p$,*

$$(2) \quad \operatorname{im} T^t = J^t, \quad \dim_{\mathbb{F}_p} J^t = p - t, \quad |\operatorname{im} T^t| = p^{p-t}.$$

In particular $T^p = 0$, and this exponent is sharp.

(ii) *If $1 \leq t \leq p$ and $g \in \mathcal{V}$, then*

$$(3) \quad |(T^t)^{-1}(g)| = \begin{cases} p^p - (p^{p-t} - 1)(p-1)^t, & g = 0, \\ (p-1)^t, & 0 \neq g \in J^t, \\ 0, & g \notin J^t. \end{cases}$$

At $t = 0$ every fibre is a singleton. For $t \geq p$, the only nonempty fibre is the zero fibre, of size p^p .

(iii) *Zero is the unique recurrent state, and the functional graph is one rooted component with its root loop retained. If $\delta(f)$ is the least t with $T^t(f) = 0$, then the number A_d of states of exact depth d is*

$$(4) \quad A_0 = 1, \quad A_d = (p-1)^{d-1} (p^{p-d} + p - 2) \quad (1 \leq d \leq p).$$

Every nonzero target in J^1 has $p-1$ immediate predecessors, every target outside J^1 has none, and zero has

$$(5) \quad p^p - (p^{p-1} - 1)(p-1)$$

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immediate predecessors, including its loop.

(iv) On the complex vector space with basis $\{e_f : f \in \mathcal{V}\}$, define the deterministic transition operator $Pe_f = e_{T(f)}$. Then

$$(6) \quad \chi_P(\lambda) = (\lambda - 1)\lambda^{p^p-1}.$$

Besides one $J_1(1)$ block, the number m_s of zero-Jordan blocks of size s is

$$(7) \quad m_s = (p-1)^2 p^{p-s-1} \quad (1 \leq s < p), \quad m_p = p-1.$$

2. THE DIFFERENCE FLAG AND ANCHORED LIFTING

For $0 \leq j < p$, regard

$$e_j(x) = \binom{x}{j} = \frac{x(x-1)\cdots(x-j+1)}{j!}$$

as a function on $\mathbb{F}_p$, with $e_0 = 1$. Their evaluation matrix on $0, 1, \dots, p-1$ is lower triangular with diagonal one, so $e_0, \dots, e_{p-1}$ form a basis of $\mathcal{V}$. Pascal's identity gives

$$(8) \quad Ne_0 = 0, \quad Ne_j = e_{j-1} \quad (1 \leq j < p).$$

Consequently

$$(9) \quad J^i = \text{span}_{\mathbb{F}_p}\{e_0, \dots, e_{p-1-i}\}, \quad \dim J^i = p-i.$$

This is the familiar one-block cyclic-difference flag.

Lemma 2 (nonzero directions and zero anchoring). *Let $a \in \mathbb{F}_p^\times$ and $0 \leq i < p$. The restriction*

$$D_a : J^i \longrightarrow J^{i+1}$$

is onto and has kernel $\mathbb{F}_p e_0$. For each $h \in J^{i+1}$ and each $b \in \mathbb{F}_p$, exactly one solution $f \in J^i$ of $D_a f = h$ satisfies $f(0) = b$.

Proof. Choose the integer representative $r \in \{1, \dots, p-1\}$ of a . Since $\tau_a = \tau^r = (I + N)^r$,

$$D_a = NU_a(N), \quad U_a(z) = \sum_{k=1}^r \binom{r}{k} z^{k-1}, \quad U_a(0) = a.$$

The polynomial $U_a(N)$ is invertible because N is nilpotent and its constant coefficient is nonzero. It preserves every J^i , commutes with N , and acts on $\ker N = \mathbb{F}_p e_0$ as multiplication by a . Equations (8)–(9) now show that D_a maps J^i onto J^{i+1} with kernel $\mathbb{F}_p e_0$. A nonempty solution set is therefore an affine line $f_0 + \mathbb{F}_p e_0$. Since $e_0(0) = 1$, evaluation at zero is a bijection from this line to $\mathbb{F}_p$, proving the anchored assertion. $\square$

Proof of Theorem 1(i)–(ii). Suppose a trajectory has not yet reached zero and record its direction word

$$(a_0, \dots, a_{t-1}) = (f_0(0), \dots, f_{t-1}(0)) \in (\mathbb{F}_p^\times)^t.$$

Each nonzero difference raises the flag index by one, while a zero direction sends the state to zero. Hence $T^t(\mathcal{V}) \subseteq J^t$ and $T^p = 0$.

Conversely, fix $0 \neq g \in J^t$ and a word $(a_0, \dots, a_{t-1}) \in (\mathbb{F}_p^\times)^t$. Set $f_t = g$ and solve backward, for $i = t-1, \dots, 0$,

$$(10) \quad D_{a_i} f_i = f_{i+1}, \quad f_i(0) = a_i, \quad f_i \in J^i.$$

Lemma 2 gives exactly one f_i at every step. Distinct words give distinct sources because a forward orbit recovers each a_i . Thus every nonzero $g \in J^t$ has exactly $(p-1)^t$ sources, proving the second line of (3) and the reverse inclusion in (2). A target outside J^t is unreachable. Subtracting the mass of the $p^{p-t} - 1$ nonzero targets from the p^p sources yields the zero line of (3).

For sharpness, let $f_\star = \sum_{j=0}^{p-1} e_j$. Repeated use of (8) gives

$$N^k f_\star = \sum_{j=0}^{p-1-k} e_j, \quad (N^k f_\star)(0) = 1 \quad (0 \leq k < p).$$

The selected direction therefore remains one until the last step: $T^{p-1} f_\star = e_0 \neq 0$ and $T^p f_\star = 0$. $\square$

3. THE ROOTED GRAPH AND TRANSITION OPERATOR

Proof of Theorem 1(iii). The relation $T^p = 0$ makes zero the sole recurrent state. Formula (3) at $t = 1$ gives every immediate indegree and (5). Thus the graph is a single rooted component; the stated indegrees describe all of its branching.

Let $Z_t = |(T^t)^{-1}(0)|$. Exact depth d means membership in the d th zero fibre but not the $(d - 1)$ st, so $A_d = Z_d - Z_{d-1}$. Using (3), with $Z_0 = 1$ and $Z_p = p^p$, gives

$$Z_d - Z_{d-1} = (p - 1)^{d-1} (p^{p-d} + p - 2)$$

for every $1 \leq d \leq p$, including the two endpoints. $\square$

Proof of Theorem 1(iv). Put $E = P^p$. Since $T^p(f) = 0$ for every state, $Ee_f = e_0$; hence E is a rank-one idempotent commuting with P . This gives the invariant decomposition

$$\mathbb{C}^V = \text{im } E \oplus \ker E.$$

The restriction of P to $\text{im } E = \mathbb{C}e_0$ is the identity, whereas $Q = P|_{\ker E}$ is nilpotent of exponent at most p . This proves (6).

For $0 \leq t \leq p$, the image of P^t is spanned by $\{e_g : g \in J^t\}$ and therefore has dimension p^{p-t} . Moreover

$$P^t(\ker E) = \text{im } P^t \cap \ker E.$$

Indeed, only the reverse inclusion needs comment: if $y = P^t x$ and $Ey = 0$, then $P^t(x - Ex) = y$. Since E has rank one on $\text{im } P^t$, it follows that

$$(11) \quad \text{rank } Q^t = p^{p-t} - 1 =: r_t.$$

For a nilpotent operator, the number of blocks of exact size $s < p$ is the second rank difference

$$m_s = r_{s-1} - 2r_s + r_{s+1},$$

while $m_p = r_{p-1}$. Substitution into (11) gives (7). This rank-to-block conversion is generic linear algebra and is not part of the claimed mechanism. $\square$

Remark 3 (the smallest boundary). At $p = 2$, the image sizes are 4, 2, 1, the depth layers are 1, 2, 1, and the transition operator has $J_1(1) \oplus J_1(0) \oplus J_2(0)$. Thus the sharp clock and the top zero block already occur in the first admissible prime.

4. EXACT CONTROLS, OWNERSHIP, AND LIMITS

The paper-local verifier represents functions directly as value tuples and starts from the literal update (1). It exhausts every state, every target, and every time through $p = 5$, including the direction-word lift behind (10). A separate modular-matrix certificate checks the binomial flag, all ranks, and anchor injectivity for $p = 2, 3, 5, 7, 11, 13, 17, 19$. The settled run makes 44,689 exact assertions over 3,156 literal arrows. These computations pressure-test arithmetic and boundary cases; they do not prove the theorem.

The ownership subtraction is intentionally severe. The operators $D_a = \tau_a - I$, augmentation powers, and their nilpotency belong to the finite-difference background represented by [1]. Fixed linear finite dynamical systems and their nilpotent flags are also established territory [2]. Internally, the A05 scout already discarded iteration of one fixed cyclic difference, and P164 already used a nonlinear first step followed by a fixed cyclic-difference tail. Their image flags, kernel cosets, clocks, and Jordan conversions all receive zero separation credit here.

What remains is narrower: (1) selects its direction at every epoch, the nonzero direction word is observable along the orbit, and the same observed values anchor each reverse integration uniquely. That feedback mechanism yields the simultaneous nonuniform zero/nonzero fibre atlas (3) and the $p - 1$ top Jordan blocks. A bounded exact-literal search found no direct owner of this conjunction, but a search nonhit does not establish novelty, priority, or freedom to operate. The note therefore remains OWNER_THIN and HOLD_EXTERNAL.

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